

Tabla de Ensamblado AVR - Problema 3a

Dirección	Hexadecimal	Binario (Instrucción)	Mnemotécnico	Direccionamiento	SREG
0x0000	E000	(1110 0000 0000 0000) ₂	ldi r16, \$0	Inmediato	-
0x0001	2E00	(0010 1110 0000 0000) ₂	mov r0, r16	Directo a registro	-
0x0002	EF0F	(1110 1111 0000 1111) ₂	ldi r16, \$ff	Inmediato	-
0x0003	2E10	(0010 1110 0001 0000) ₂	mov r1, r16	Directo a registro	-
0x0004	E0D2	(1110 0000 1101 0010) ₂	ldi yh, \$02	Inmediato	-
0x0005	E0C1	(1110 0000 1100 0001) ₂	ldi yl, \$01	Inmediato	-
0x0006	E008	(1110 0000 0000 1000) ₂	ldi r16, \$08	Inmediato	-
0x0007	9300 0200	(1001 0011 0000 0000 0000 0010 0000 0000) ₂	sts \$200, r16	Directo a memoria	-
0x0009	E003	(1110 0000 0000 0011) ₂	ldi r16, \$03	Inmediato	-
0x000A	9300 0201	(1001 0011 0000 0000 0000 0010 0000 0001) ₂	sts \$201, r16	Directo a memoria	-
0x000C	9110 0202	(1001 0001 0001 0000 0000 0010 0000 0010) ₂	lds r17, \$202	Directo a memoria	-
0x000E	2E01	(0010 1110 0000 0001) ₂	mov r0, r17	Directo a registro	-
0x000F	010E	(0000 0001 0000 1110) ₂	movw r0, y	Directo a registro	-
0x0010	9009	(1001 0000 0000 1001) ₂	ld r0, y+	Indirecto post-inc.	-
0x0011	900A	(1001 0000 0000 1010) ₂	ld r0, -y	Indirecto pre-dec.	-
0x0012	800A	(1000 0000 0000 1010) ₂	ldd r0, y+2	Indirecto c/ desplazam.	-

Desensamblado AVR - Problema 3b

Secuencia de instrucciones expresada en hexa:

40 E1 55 E1 64 2F 45 17 2C F4 43 95 00 00 00 00 0C 94 03 00 00 00 98 95

Teniendo en cuenta que AVR es Little Endian, (los pares se leen de derecha a izquierda), el código es el siguiente:

Hexa	Binario	Instrucción
E1 40	(1110 0001 0100 0000) ₂	ldi r20, \$10
E1 55	(1110 0001 0101 0101) ₂	ldi r21, \$15
2F 64	(0010 1111 0110 0100) ₂	mov r22, r20
17 45	(0001 0111 0100 0101) ₂	cp r20, r21
F4 2C	(1111 0100 0010 1100) ₂	brge 5
95 43	(1001 0101 0100 0011) ₂	inc r20
00 00	(0000 0000 0000 0000) ₂	nop
00 00	(0000 0000 0000 0000) ₂	nop
94 0C 00 03	(1001 0100 0000 1100 ... 0011) ₂	jmp \$0003
00 00	(0000 0000 0000 0000) ₂	nop
95 98	(1001 0101 1001 1000) ₂	break

Anexo: Referencias del Manual de Instrucciones (Atmel AVR)

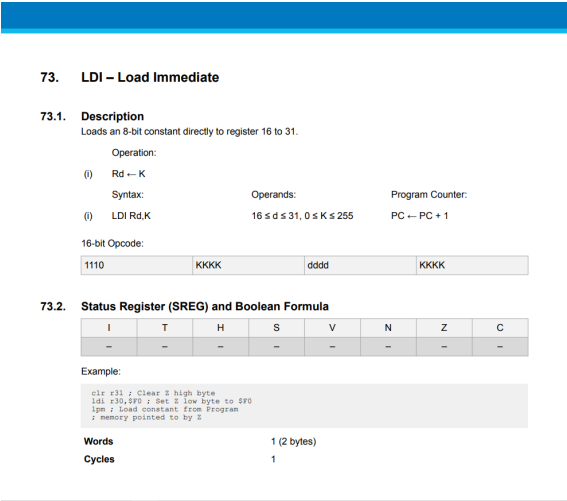


Figure 1: Instrucción LDI: Carga inmediata de 8 bits.

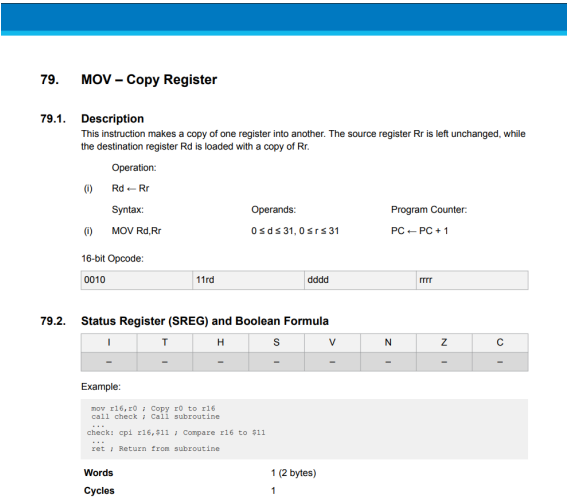


Figure 2: Instrucción MOV: Copia entre registros.

121. STS – Store Direct to Data Space

121.1. Description

Stores one byte from a Register to the data space. For parts with SRAM, the data space consists of the Register File, I/O memory, and internal SRAM (and external SRAM if applicable). For parts without SRAM, the data space consists of the Register File only. The EEPROM has a separate address space.

A 16-bit address must be supplied. Memory access is limited to the current data segment of 64KB. The STS instruction uses the RAMPD Register to access memory above 64KB. To access another data segment in devices with more than 64KB data space, the RAMPD in register in the I/O area has to be changed.

This instruction is not available in all devices. Refer to the device specific instruction set summary.

Operation:

(i) $(k) \leftarrow Rr$

Syntax:

Operands:

Program Counter:

(i) STS k,Rr

$0 \leq r \leq 31, 0 \leq k \leq 65535$

$PC \leftarrow PC + 2$

32-bit Opcode:

1001	001d	dddd	0000
kkkk	kkkk	kkkk	kkkk

121.2. Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	–	–	–	–	–

Example:

lda r2,\$FF00 ; Load r2 with the contents of data space location \$FF00

Figure 3: Instrucción STS: Almacenamiento directo (32 bits).

74. LDS – Load Direct from Data Space

74.1. Description

Loads one byte from the data space to a register. For parts with SRAM, the data space consists of the Register File, I/O memory, and internal SRAM (and external SRAM if applicable). For parts without SRAM, the data space consists of the register file only. The EEPROM has a separate address space.

A 16-bit address must be supplied. Memory access is limited to the current data segment of 64KB. The LDS instruction uses the RAMPD Register to access memory above 64KB. To access another data segment in devices with more than 64KB data space, the RAMPD in register in the I/O area has to be changed.

This instruction is not available in all devices. Refer to the device specific instruction set summary.

Operation:

(i) $Rd \leftarrow (k)$

Syntax:

Operands:

Program Counter:

(i) LDS Rd,k

$0 \leq d \leq 31, 0 \leq k \leq 65535$

$PC \leftarrow PC + 2$

32-bit Opcode:

1001	000d	dddd	0000
kkkk	kkkk	kkkk	kkkk

74.2. Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	–	–	–	–	–

Example:

lda r2,\$FF00 ; Load r2 with the contents of data space location \$FF00

Figure 4: Instrucción LDS: Carga directa desde SRAM (32 bits).

80. MOVW – Copy Register Word

80.1. Description

This instruction makes a copy of one register pair into another register pair. The source register pair $Rr + 1:Rr$ is left unchanged, while the destination register pair $Rd+1:Rd$ is loaded with a copy of $Rr + 1:Rr$. This instruction is not available in all devices. Refer to the device specific instruction set summary.

Operation:

(i) $Rd+1:Rd \leftarrow Rr+1:Rr$

Syntax: Operands: Program Counter:

(i) MOVW $Rd+1:Rd, Rr+1:Rr$ $d \in \{0,2,...,30\}$, $r \in \{0,2,...,30\}$ $PC \leftarrow PC + 1$

16-bit Opcode:

0000	0001	dddd	rrrr
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80.2. Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
--	--	--	--	--	--	--	--

Figure 5: Instrucción MOVW: Copia de pares de registros (Word).

(iii) LD $Rd, -Y$ $0 \leq d \leq 31$ $PC \leftarrow PC + 1$

(iv) LDD $Rd, Y+q$ $0 \leq d \leq 31$, $0 \leq q \leq 63$ $PC \leftarrow PC + 1$

16-bit Opcode:

(i)	1000	000d	dddd	1000
(ii)	1001	000d	dddd	1001
(iii)	1001	000d	dddd	1010
(iv)	10q0	qq0d	dddd	1qqq

71.2. Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
--	--	--	--	--	--	--	--

Figure 6: Instrucción LD/LDD: Variantes con punteros indirectos.